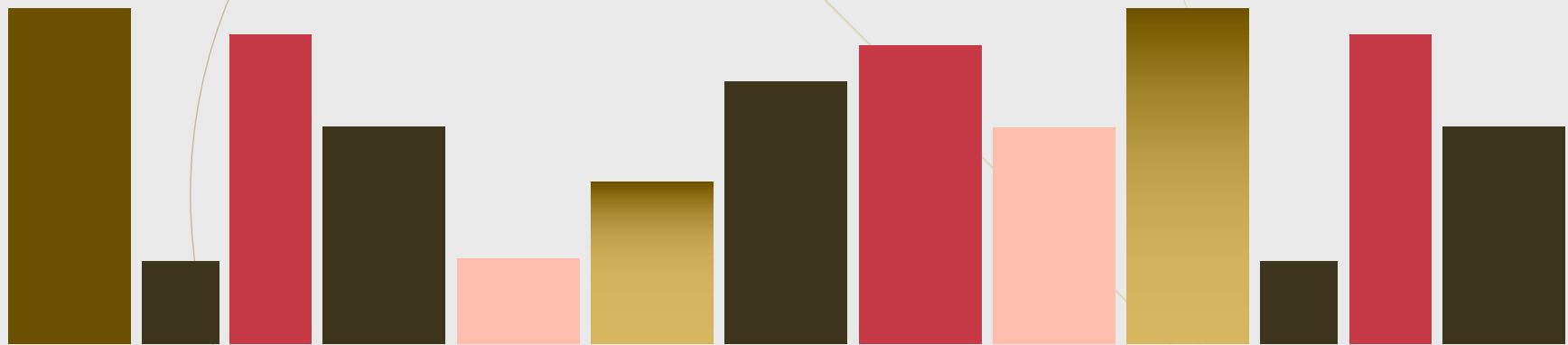


ZEN Univ. IUT-1



6th

Exercises.

1.

(a)

$$\left(\frac{219}{383} \right) = (-1)^{219 \cdot 383/4} \left(\frac{383}{219} \right)$$

$$= (-1) \left(\frac{164}{219} \right)$$

$$= (-1) \left(\frac{164}{3} \right) \left(\frac{164}{73} \right)$$

$$= (-1) \left(\frac{2}{3} \right) \left(\frac{18}{73} \right)$$

$$= (-1) (-1) \left(\frac{73}{18} \right) (-1)^{73 \cdot 18/4}$$

$$= \left(\frac{73}{2} \right) \left(\frac{73}{9} \right) = \left(\frac{1}{2} \right) \left(\frac{1}{9} \right) = 1$$

219 isn't prime,
 $\gcd(164, 219) = 1$

18 isn't prime
 $\gcd(18, 73) = 1$

383 is prime, so x has 2 solutions. ~~2~~

$$(b) \left(\frac{118}{135} \right) = \left(\frac{118}{27} \right) \left(\frac{118}{5} \right)$$

$$= \left(\frac{1}{27} \right) \left(\frac{3}{5} \right)$$

$$= 1 \cdot (-1)$$

$$= -1.$$

There're no solutions. ~~2~~

$$2. \quad x^2 \equiv 74 \pmod{203}$$

$$\Leftrightarrow \begin{cases} x^2 \equiv 4 \pmod{7} \\ x^2 \equiv 16 \pmod{29} \end{cases}$$

And $\underbrace{4}_{2^2}, \underbrace{16}_{4^2}$ is square residue, so there're $2^2 = 4$ solutions (by CRT) ~~10~~.

$$3. (a) \exists x. x^2 + 6x + 11 \equiv 0 \pmod{p}$$

$$\Leftrightarrow \exists x. (x+3)^2 + 2 \equiv 0 \pmod{p}$$

$$\Leftrightarrow \exists x. (x+3)^2 \equiv -2 \pmod{p}$$

$$\Leftrightarrow \left(\frac{-2}{p} \right) = 1$$

||

$$\left(\frac{-1}{p} \right) \left(\frac{2}{p} \right)$$

$p_{\text{mod } 8}$	$\left(\frac{-1}{p}\right)$	$\left(\frac{2}{p}\right)$	$\left(\frac{-2}{p}\right) = \left(\frac{-1}{p}\right) \left(\frac{2}{p}\right)$
1	1	1	1
3	-1	-1	1
5	1	-1	-1
7	-1	1	-1

finally,
 $\Rightarrow$

$$p \equiv 1, 3 \pmod{8} \quad \text{■}$$

3. (b) If $p \mid n^2 + 3$, $p = k(n^2 + 3)$ for some $k < p$.

$$\text{so } n^2 + 3 \equiv 0 \pmod{p}. \quad n^2 \equiv -3.$$

$$\left(\frac{-3}{p} \right) = 1$$

$$\left(\frac{-1}{p} \right) \left(\frac{3}{p} \right)$$

$$= (-1)^{\frac{p-1}{2}} (-1)^{2 \cdot (p-1)/4} \cdot \left(\frac{p}{3} \right)$$

$$= \left((-1)^{\frac{p-1}{2}} \right)^2 \cdot \left(\frac{p}{3} \right) = \left(\frac{p}{3} \right)$$

$$\left(\frac{p}{3}\right) = 1 \quad \Leftrightarrow \quad \underline{p \equiv 1 \pmod{3}} \quad \blacksquare$$

5. Assume $2^n + 1 \mid 3^{n+1} + 1$.

For some k , $3^{n+1} + 1 = k(2^n + 1)$

Take mod 3 both sides, $1 \equiv k((-1)^n + 1)$.

If n is even, $1 \equiv 0$ which cannot be satisfied.

So n is odd.

For some m , $n = 2m + 1$.

Take some prime factor $p \mid 2^{2m} + 1$

$$2^{2m} + 1 \equiv 0 \pmod{p}$$

$$(2^m)^2 \equiv -1$$

$$\text{So } \left(\frac{-1}{p} \right) = 1.$$

Also, $p \mid 3^{2m+1} + 1$.

$$3^{2m+1} + 1 \equiv 0 \pmod{p}$$

$$3 \cdot 3^{2m} \equiv -1 \equiv 2^{2m}$$

$$\hookrightarrow 3 \equiv (3^{-m} \cdot 2^m)^2 \quad \text{so} \quad \underline{\left(\frac{3}{p}\right) = 1}$$

However, $2^{2m+1} \equiv 2 \pmod{3}$.

So we can take some p s.t. $p \equiv 2 \pmod{3}$.

With such p , $\left(\frac{3}{p}\right) = \underbrace{(-1)^{2 \cdot (p-1)/4}}_{= \underbrace{\left(\frac{-1}{p}\right) = 1}} \cdot \left(\frac{p}{3}\right)$

$$= \left(\frac{p}{3}\right)$$

$$= \underline{-1} \quad \# \quad \text{shaded circle}$$